

Accounting Fraud Analysis

of 300159.SZ Xinyan Holding

Simple, Complex, and Cost-Inflating Fraud Patterns

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PART 01

Background and Fraud Overview

This section sets the context for the case and focuses on the fraud period from 2017 to 2019.

Company Background

Xinyan Holding (300159.SZ)

- Originally focused on agricultural and pastoral machinery
- Later expanded into aviation-related operations
- Mingri Aviation consolidated from 1 November 2015





Fraud Overview

2017-2018:

- Fictitious sales contracts
- Overstated revenue
- Inflated accounts receivable

2019:

- Premature revenue and cost recognition
- Inflated accounts receivable

Overall effect: overstated revenue and accounts receivable



PART 02

Simple Fraud Pattern

This section examines whether inflated sales are reflected in abnormal accounts receivable patterns.

1. Growth in Accounts Receivable VS. Growth in Sales

Growth in Accounts Receivable (AR) = (AR in year t / AR in year t-1) – 1

Growth in Sales = (Sales in year t / Sales in year t-1) – 1

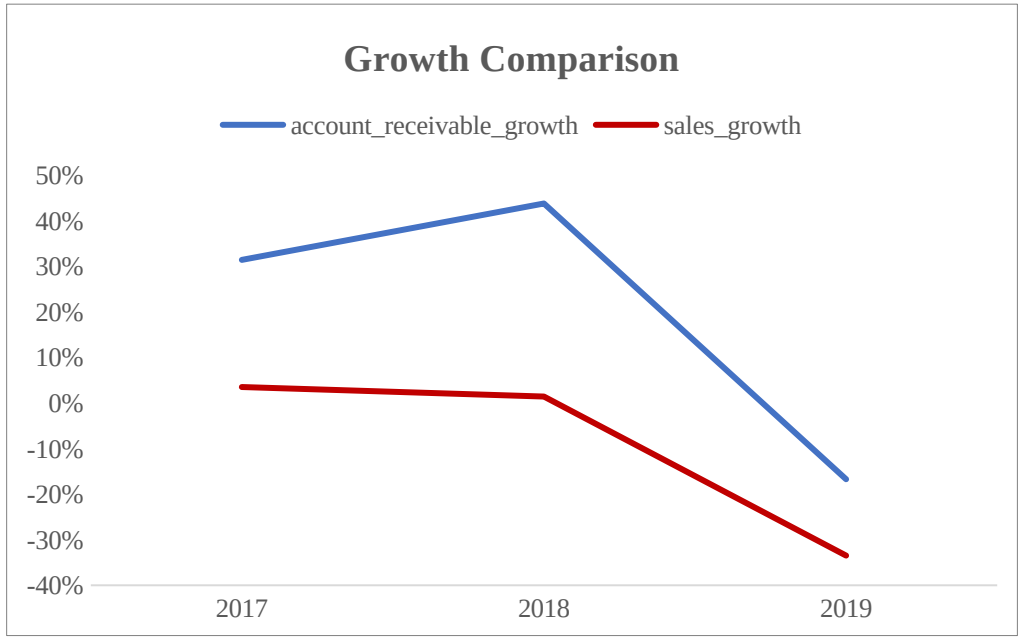


Figure description:

- AR growth > Sales growth
- Especially clear in **2017** and **2018**
- 2018: AR **+43.8%** vs Sales **+1.4%**

Interpretation:

- Inflated sales may be recorded as **AR**
- No corresponding cash collection
- AR grows faster than Sales → **abnormal**
- **Initial support** for the simple fraud pattern

2. Compare Accounts Receivable Turnover Ratio with Gross Margin

Accounts Receivable Turnover Ratio ≈ Sales / Average Accounts Receivable

Gross Margin = (Sales - Cost of goods sold) / Sales

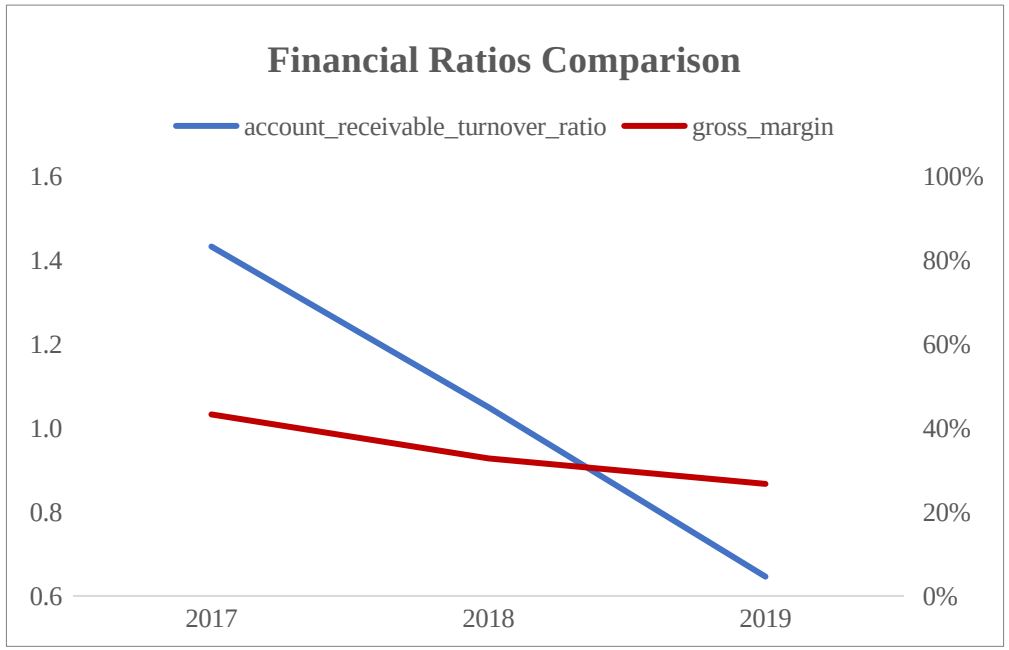


Figure description:

- AR turnover declined sharply
- **2017-2019: 1.43 → 0.65**
- Gross margin also declined
- Trend divergence was **limited**

Interpretation:

- Lower AR turnover → slower collection
- Receivable quality may be weakening
- Provides **moderate support** for the simple fraud pattern
- **Not the strongest** evidence on its own

3. Check the ratio of Net Cash Flow from operations over Net Profits

Net Cash_Profit ratio = Net Cash Flow from operations / Net Profits

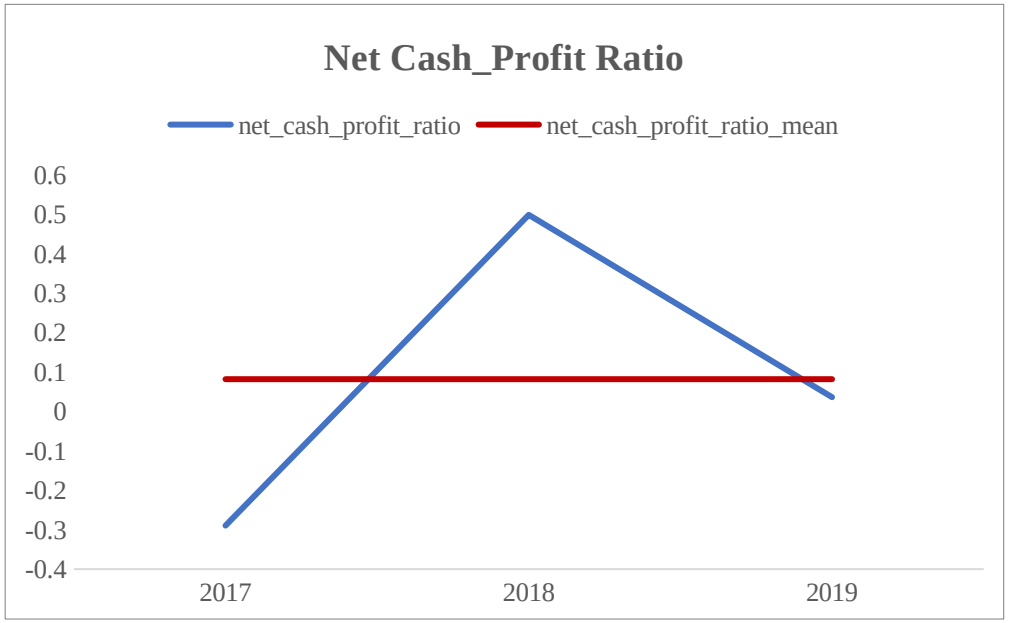


Figure description:

- Net Cash_Profit ratio remained weak in **2017-2019**
- Mean value was **around 0.08**
- **Significantly below 1**
- Cash-profit conversion was **very weak**

Interpretation:

- Reported profits were **not supported** by operating cash flow
- Inflated sales may increase profit, but **not real cash inflow**
- Net profit was **not converted into cash**
- **Strong support** for the simple fraud pattern

4. Check Accounts Receivable Bad Debt ratio

Accounts Receivable Bad Debt ratio = Allowance for Doubtful Accounts / Total Accounts Receivable

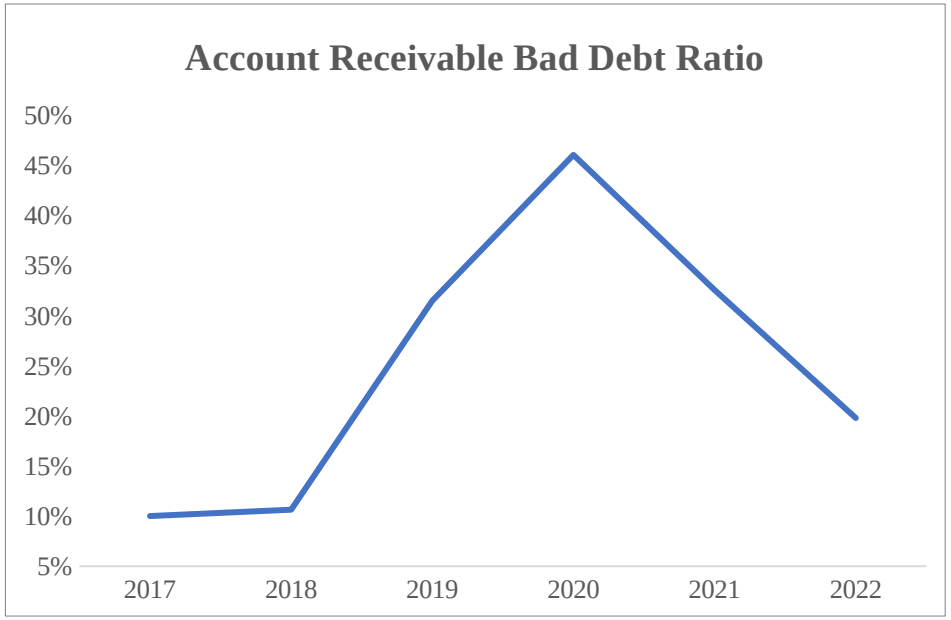


Figure description:

- AR bad debt ratio increased later
- **2017-2020: 10% → 46%**
- The rise was significant
- This is a **lagging indicator**

Interpretation:

- Fake receivables may not be impaired immediately
- Higher provisions appeared in later years
- Receivable risk increased significantly
- **Strong support** for the simple fraud pattern

Conclusion

The **simple fraud pattern (inflating Accounts Receivable)** of inflating Sales **exists**.

- **AR growth > Sales growth** → abnormal
- **AR Turnover Ratio** → declined significantly
- **Mean Net Cash_Profit ratio** → far below 1
- **AR bad debt ratio** → increased later



PART 03

Complex Fraud Pattern

This section explores whether cash recycle through outside parties supports inflated sales.

5 (1). Check the growth in Other Receivable

Growth in Other Receivable (OR) = (OR in year t / OR in year t-1) – 1

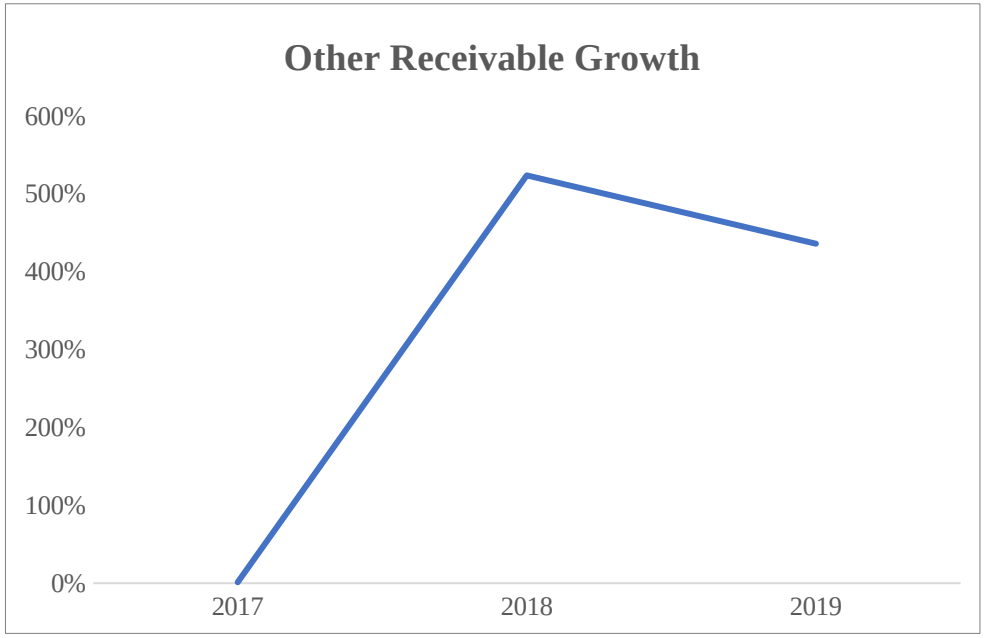


Figure description:

- OR growth increased dramatically
- Especially obvious in **2018** and **2019**
- **2018**: OR growth ≈ **523%**
- The increase was **highly abnormal**

Interpretation:

- Cash outflow may be recorded as **Other Receivable**
- Significant OR growth → **abnormal**
- May indicate fund transfers through **outside parties**
- **Strong support** for the complex fraud pattern

5 (2). Check the Other Receivable Turnover Ratio

Other Receivable Turnover Ratio = Sales / Average Other Receivable

Average Other Receivable = (Previous year + Current year Other Receivable) / 2

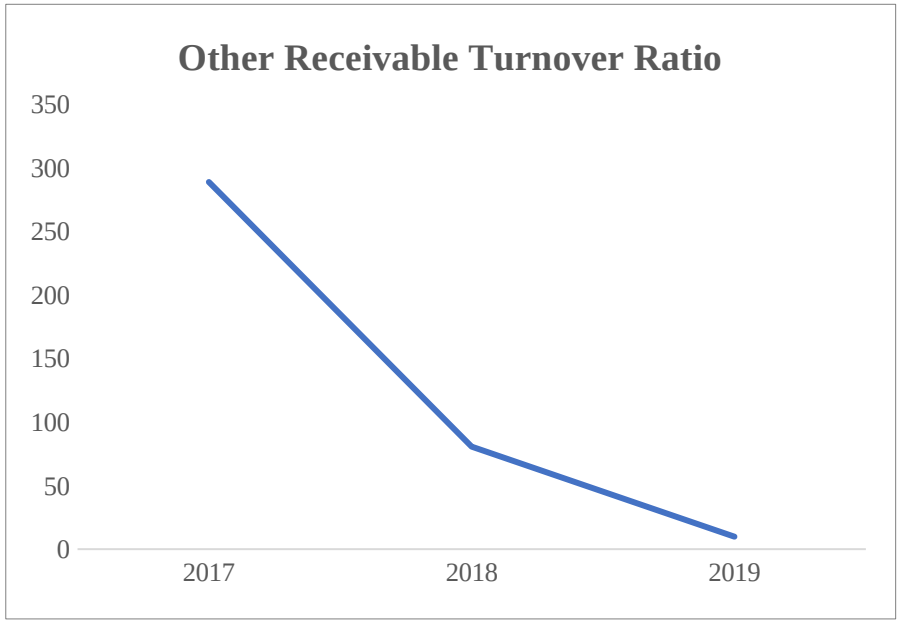


Figure description:

- OR turnover ratio declined sharply
- **2017-2019: 288.6 → 9.8**
- The decline was **very dramatic**
- Indicates rapid OR accumulation

Interpretation:

- Cash flow may accumulate in **Other Receivable**
- Falling OR turnover ratio → **abnormal**
- May indicate hidden fund transfers through **outside parties**
- **Strong support** for the complex fraud pattern

6 (1). Check the growth in Prepayment

Growth in Prepayment = (Prepayment in year t / Prepayment in year t-1) – 1

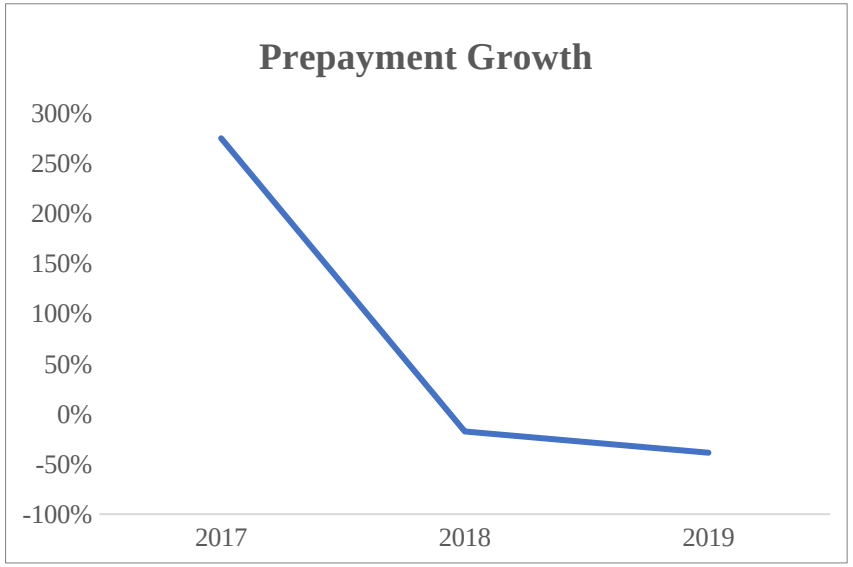


Figure description:

- Prepayment growth spiked in **2017**
- Turned **negative** in **2018** and **2019**
- The pattern **was not consistently strong**

Interpretation:

- Cash outflow may be recorded as **Prepayment**
- A spike in Prepayment may indicate **advance cash transfer**
- **Limited to moderate support** for the complex fraud pattern

6 (2). Check the Prepayment Turnover Ratio

Prepayment Turnover Ratio = Cost of goods sold / Average Prepayment

Average Prepayment = (Previous year + Current year Prepayment) / 2

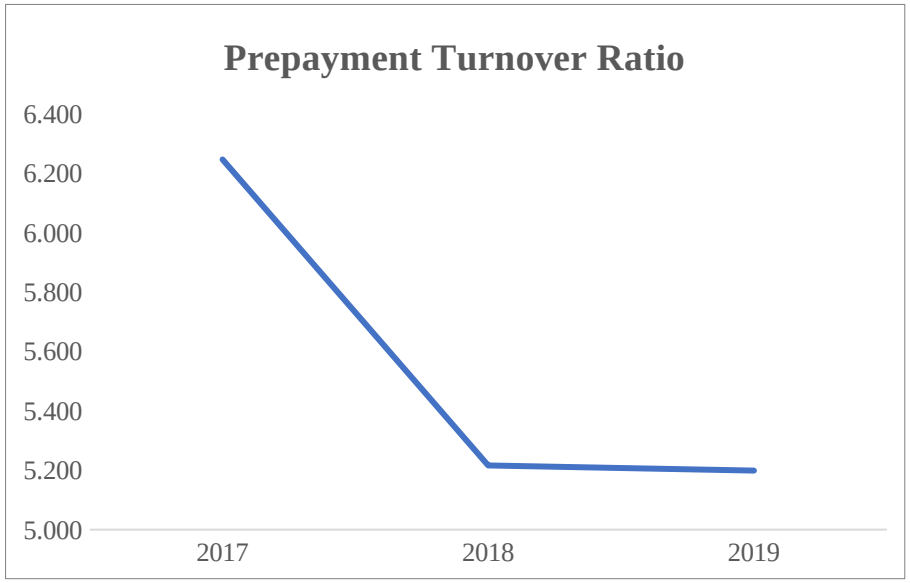


Figure description:

- Prepayment turnover ratio declined slightly
- **2017-2019: 6.25 → 5.20**
- The decline was **not very dramatic**

Interpretation:

- Lower turnover may indicate slower prepayment turnover
- The evidence was **relatively weak**
- Weaker than the **Other Receivable** evidence
- **Limited support** for the complex fraud pattern

Conclusion

The **complex fraud pattern (cash recycle)** of inflating Sales **exists**.

- **Other Receivable growth** → highly abnormal
- **Other Receivable turnover ratio** → declined sharply
- **Prepayment growth** → spiked in 2017 only
- **Prepayment turnover ratio** → weak but declining



PART 04

Cost-Inflating Fraud Pattern

This section analyzes whether abnormal accounts payable and inventory patterns suggest inflated costs.

7 (1). Growth in Account Payable VS. Growth in COGS

Growth in Account Payable (AP) = (AP in year t / AP in year t-1) – 1

Growth in COGS = (COGS in year t / COGS in year t-1) – 1

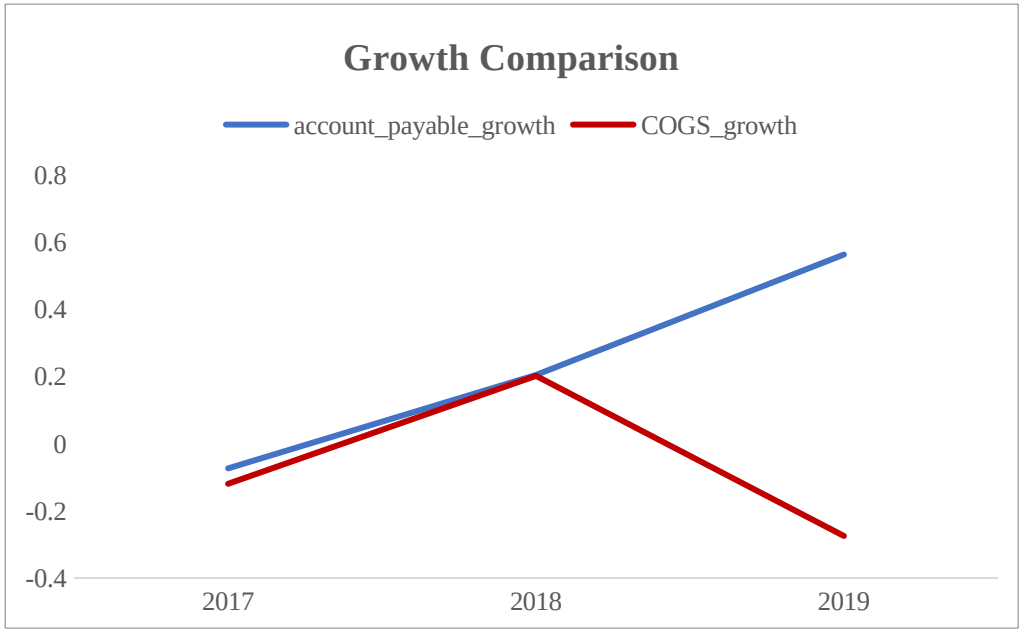


Figure description:

- AP growth exceeded COGS growth in **2017** and **2019**
- The gap was **most obvious in 2019**
- **2019: AP +56.2%** vs. **COGS -27.5%**
- The pattern was **not consistently strong**

Interpretation:

- AP grew much faster than COGS → **abnormal**
- May indicate inflated purchases recorded through **AP**
- Part of inflated purchases may be transferred to **COGS**
- **Moderate support** for the cost-inflating pattern

7 (2). Check the Account Payable Turnover Ratio

Account Payable Turnover Ratio = Cost of goods sold (COGS) / Average Account Payable

Average Account Payable = (Previous year + Current year Account Payable) / 2

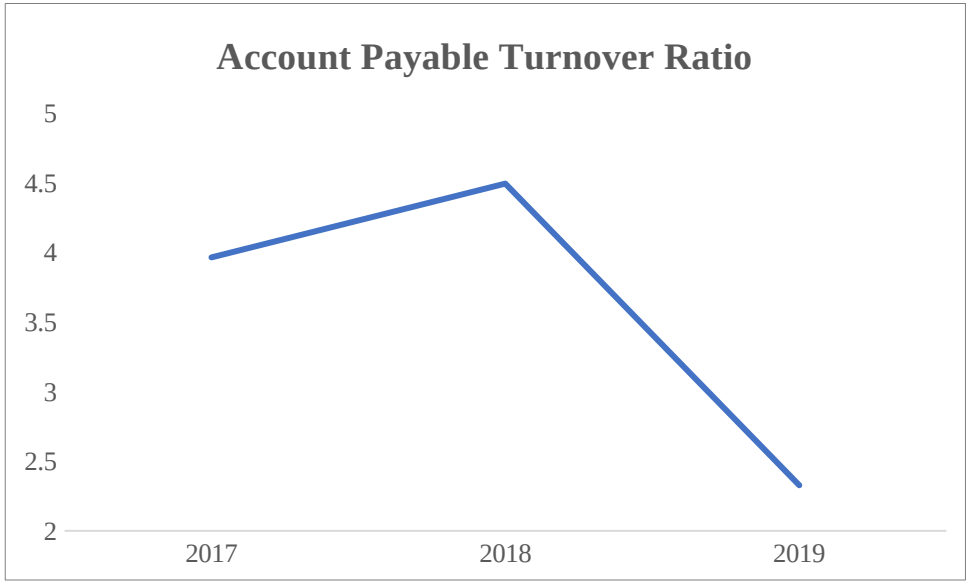


Figure description:

- AP turnover ratio fluctuated at first
- But declined clearly in the **later fraud period**
- Indicates faster **AP accumulation**

Interpretation:

- Falling AP turnover ratio → **abnormal**
- AP accumulated faster than **COGS**
- May indicate inflated purchases / inflated **COGS**
- **Limited to moderate support**

8 (1). Growth in Inventory VS. Growth in Sales

Growth in Inventory = (Inventory in year t / Inventory in year t-1) – 1

Growth in Sales = (Sales in year t / Sales in year t-1) – 1

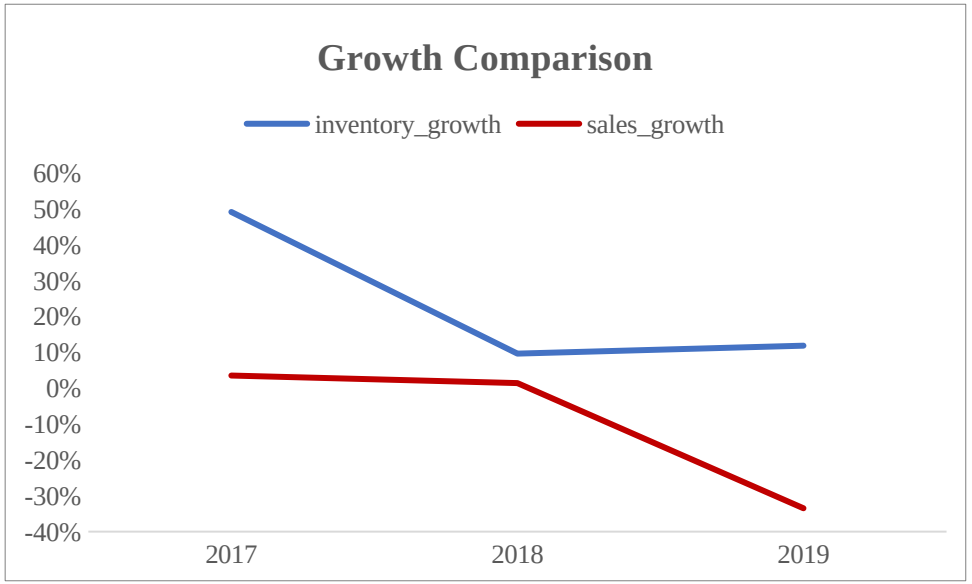


Figure description:

- Inventory growth exceeded Sales growth
- The gap became more obvious in the **later fraud period**
- Inventory accumulated abnormally during **2017-2019**
- The pattern was **strong**

Interpretation:

- Inventory grew faster than Sales → **abnormal**
- May indicate inflated ending **Inventory**
- Part of inflated purchases may be transferred to **Inventory**
- **Strong support** for the cost-inflating pattern

8 (2). Check the Inventory Turnover Ratio

Inventory Turnover Ratio = Cost of goods sold (COGS) / Average Inventory

Average Inventory = (Previous year + Current year Inventory) / 2

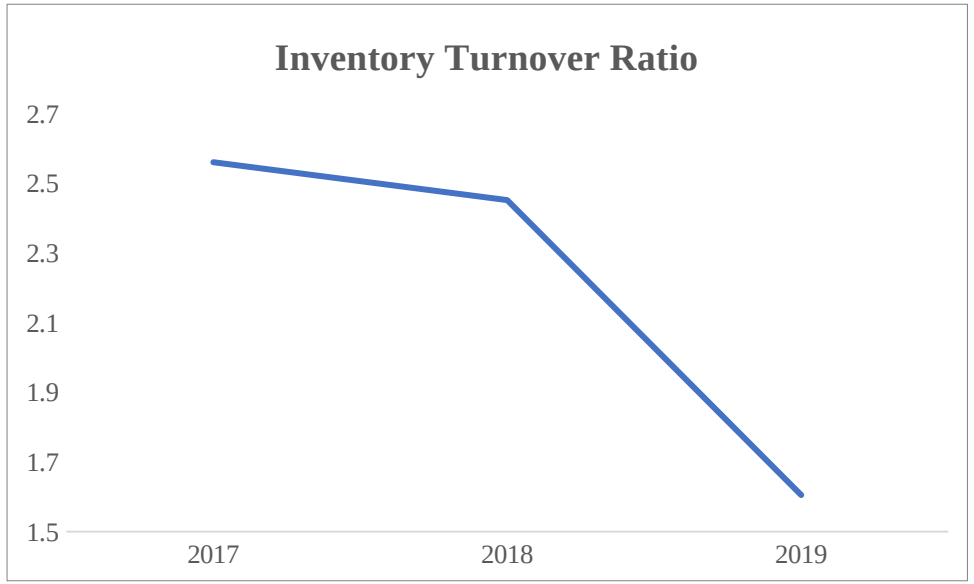


Figure description:

- Inventory turnover ratio kept declining
- **2017-2019: 2.56 → 1.60**
- The decline became more obvious later
- Indicates abnormal inventory accumulation

Interpretation:

- Falling inventory turnover → **abnormal**
- Inventory accumulated faster than **COGS**
- May indicate inflated ending **Inventory**
- **Strong support** for the cost-inflating pattern

Conclusion

The **cost-inflating fraud pattern exists**.

- **AP growth** → abnormal
- **AP turnover ratio** → weak but declining
- **Inventory growth** → abnormal
- **Inventory turnover ratio** → declined significantly



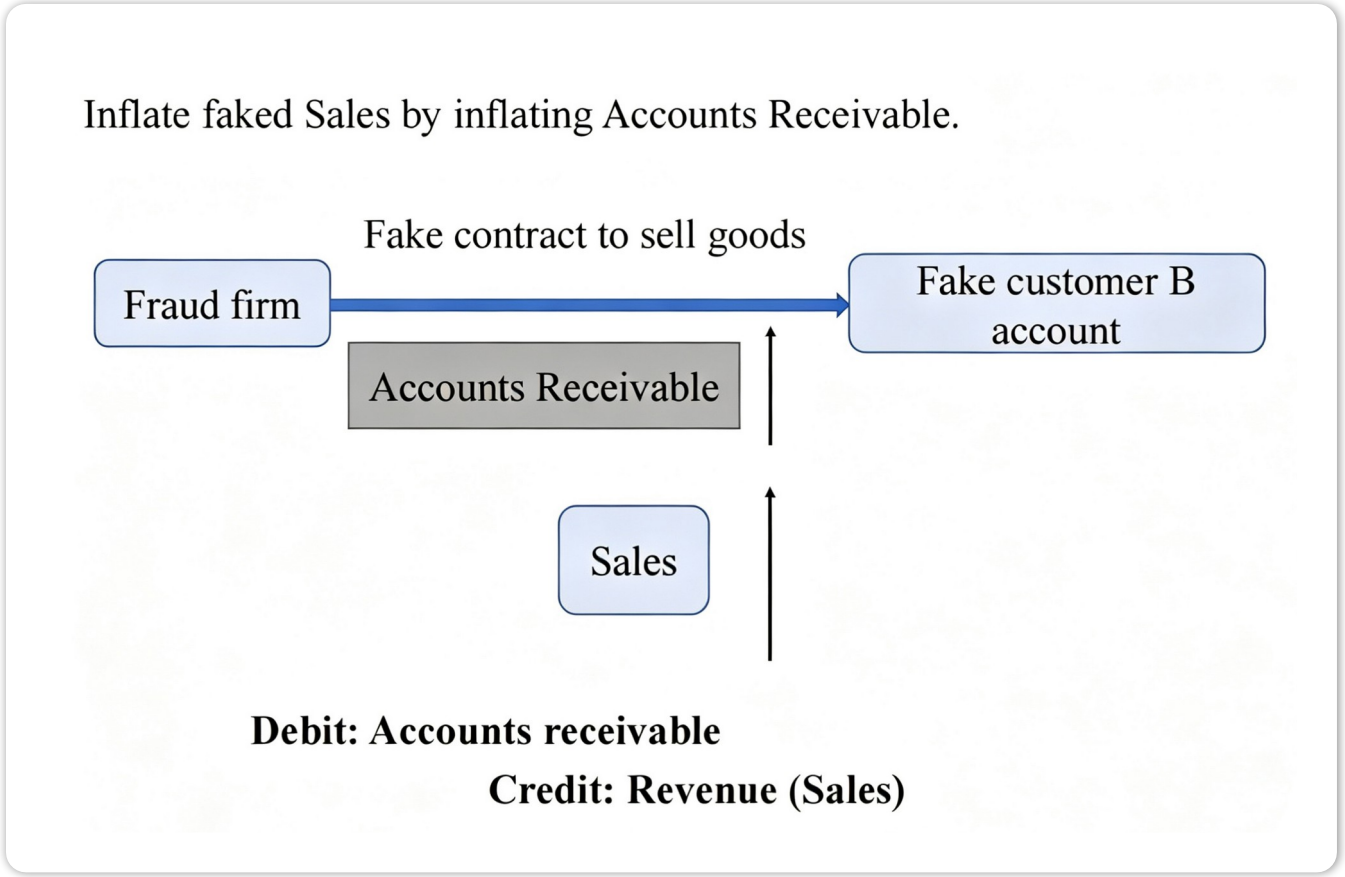
PART 05

Flow Charts, Conclusion, and References

This section presents the fraud mechanisms, final assessment, and reference sources.

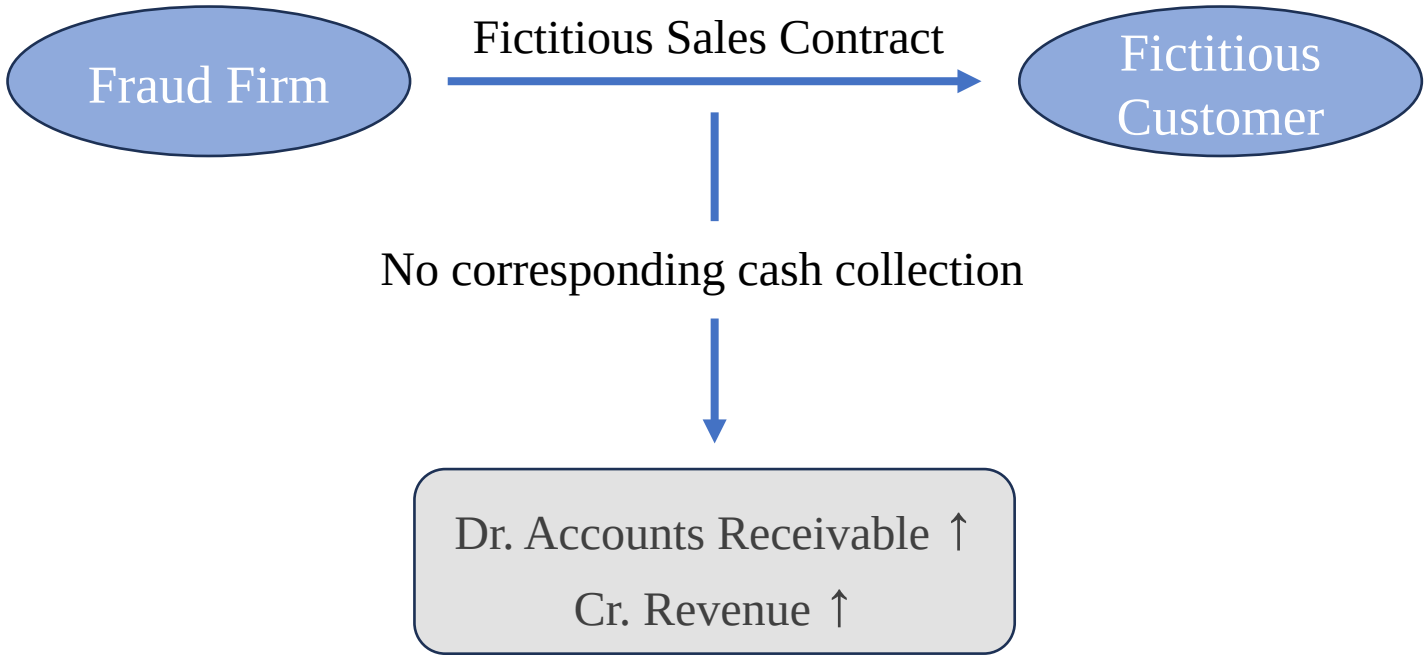
Flow Chart: Simple Fraud Pattern

Original illustration from lecture slides:



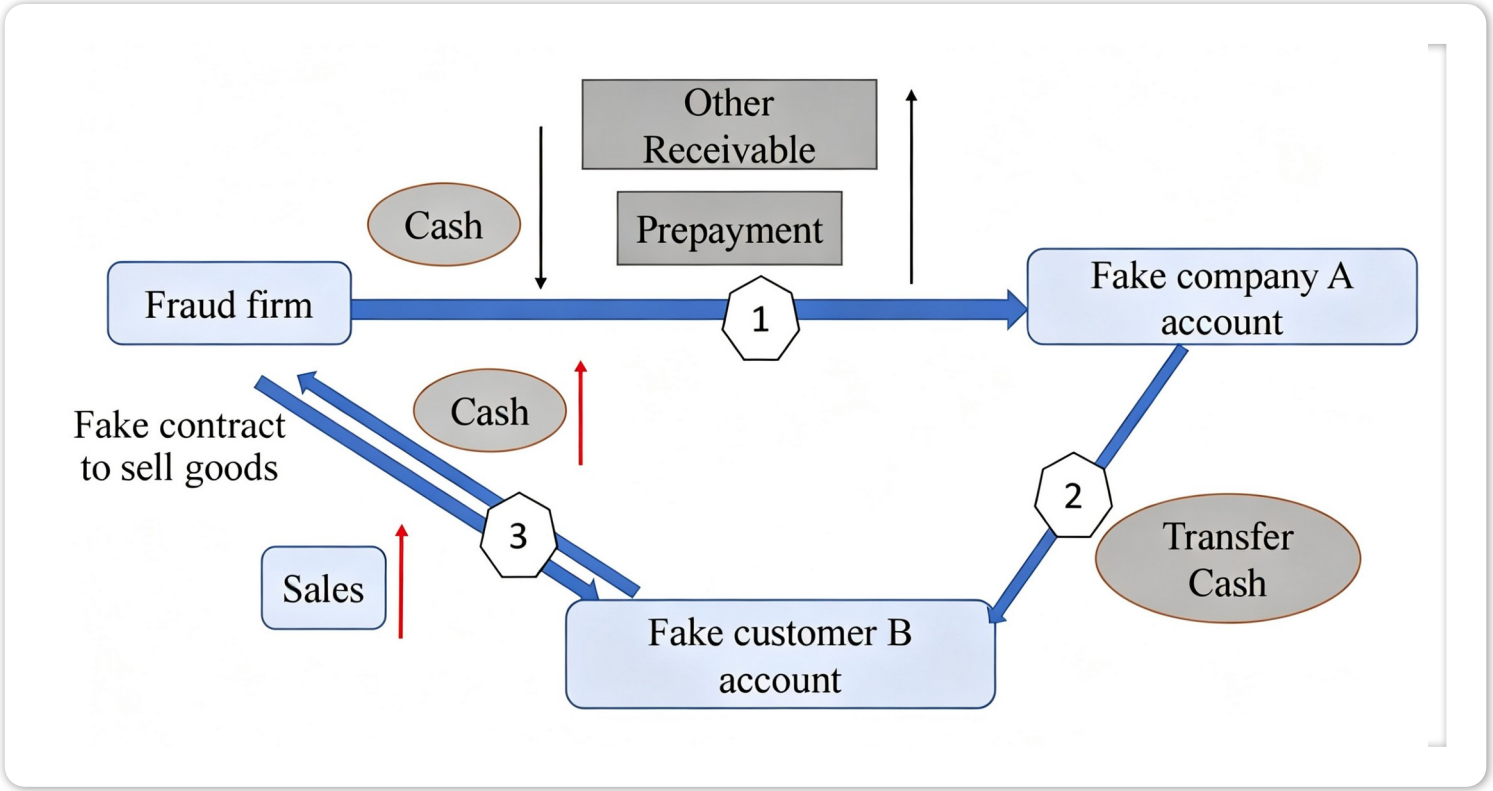
Flow Chart: Simple Fraud Pattern

Reconstructed mechanism:



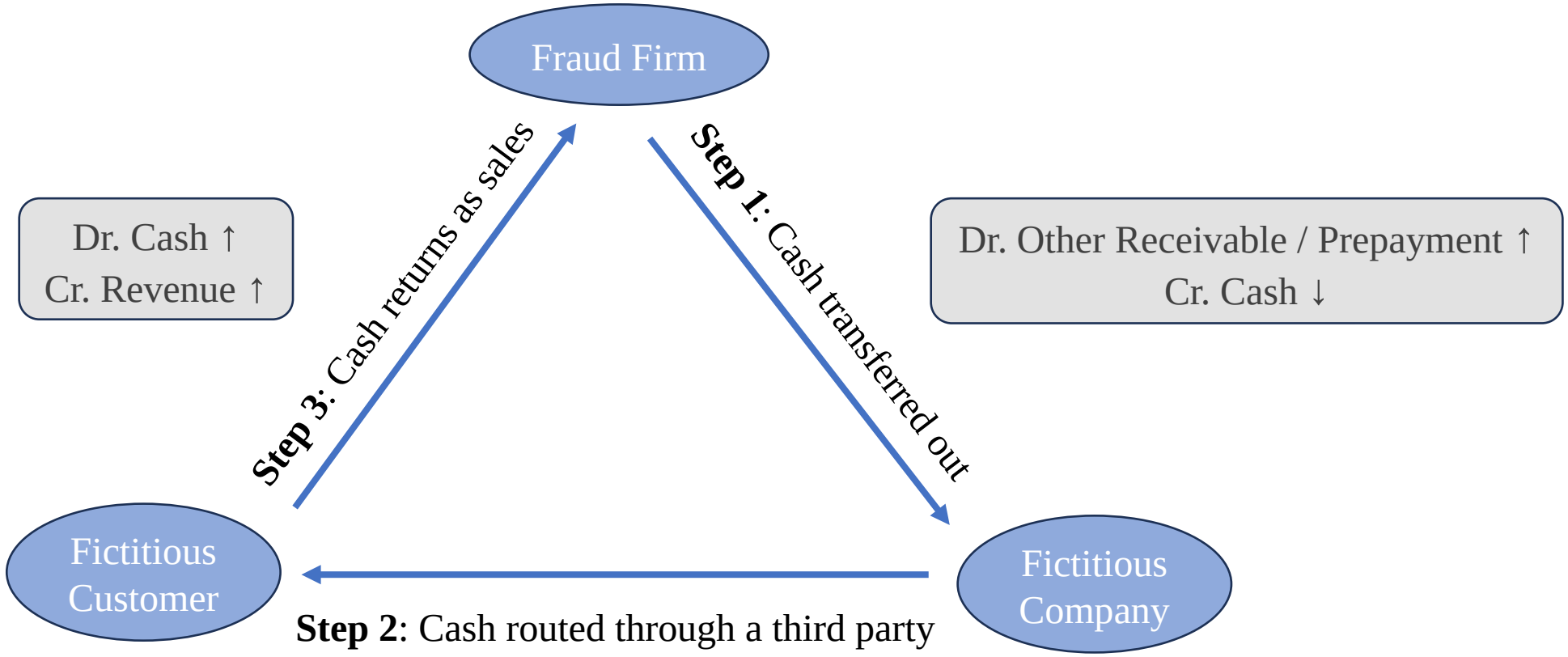
Flow Chart: Complex Fraud Pattern

Original illustration from lecture slides:



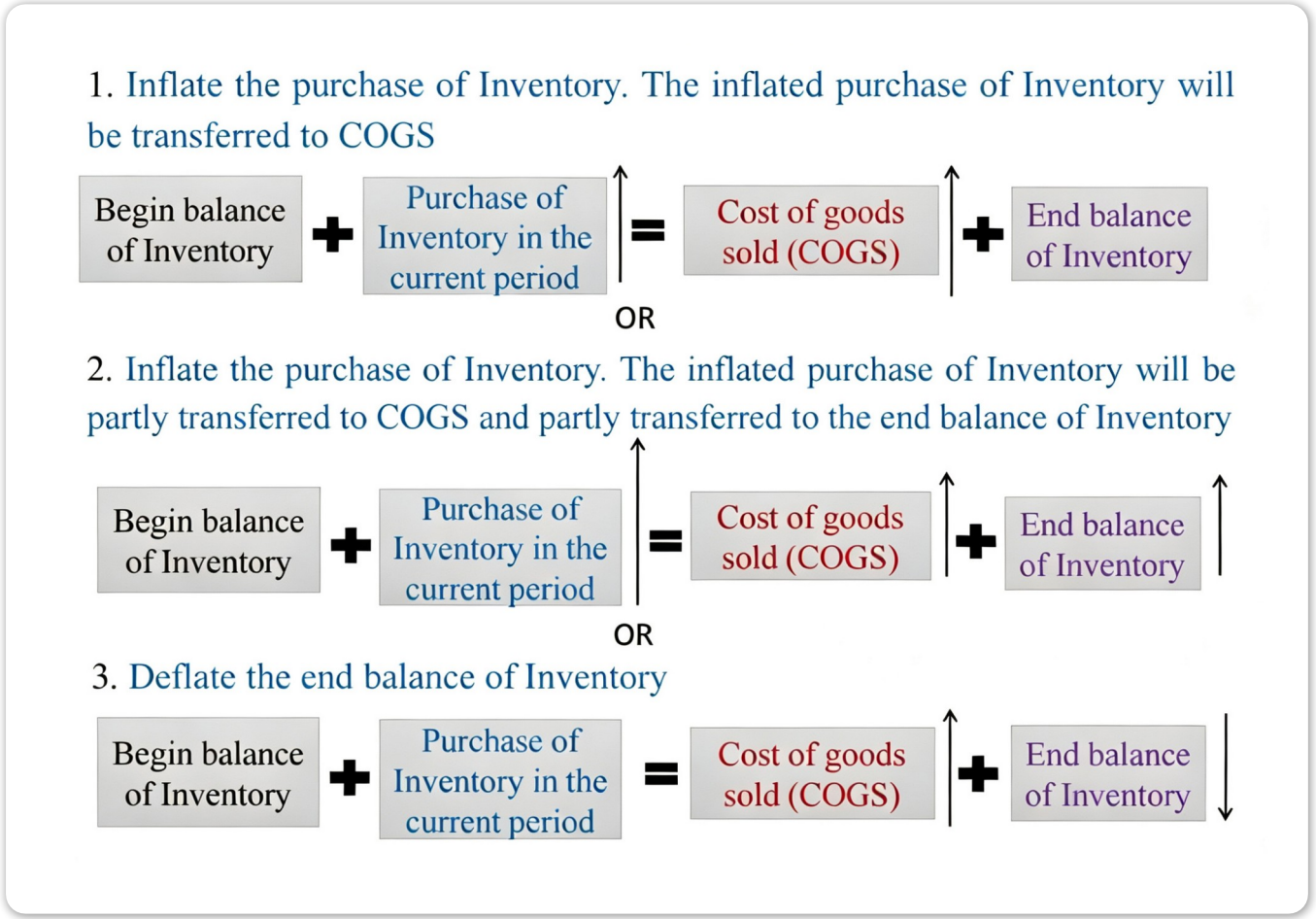
Flow Chart: Complex Fraud Pattern

Reconstructed mechanism:



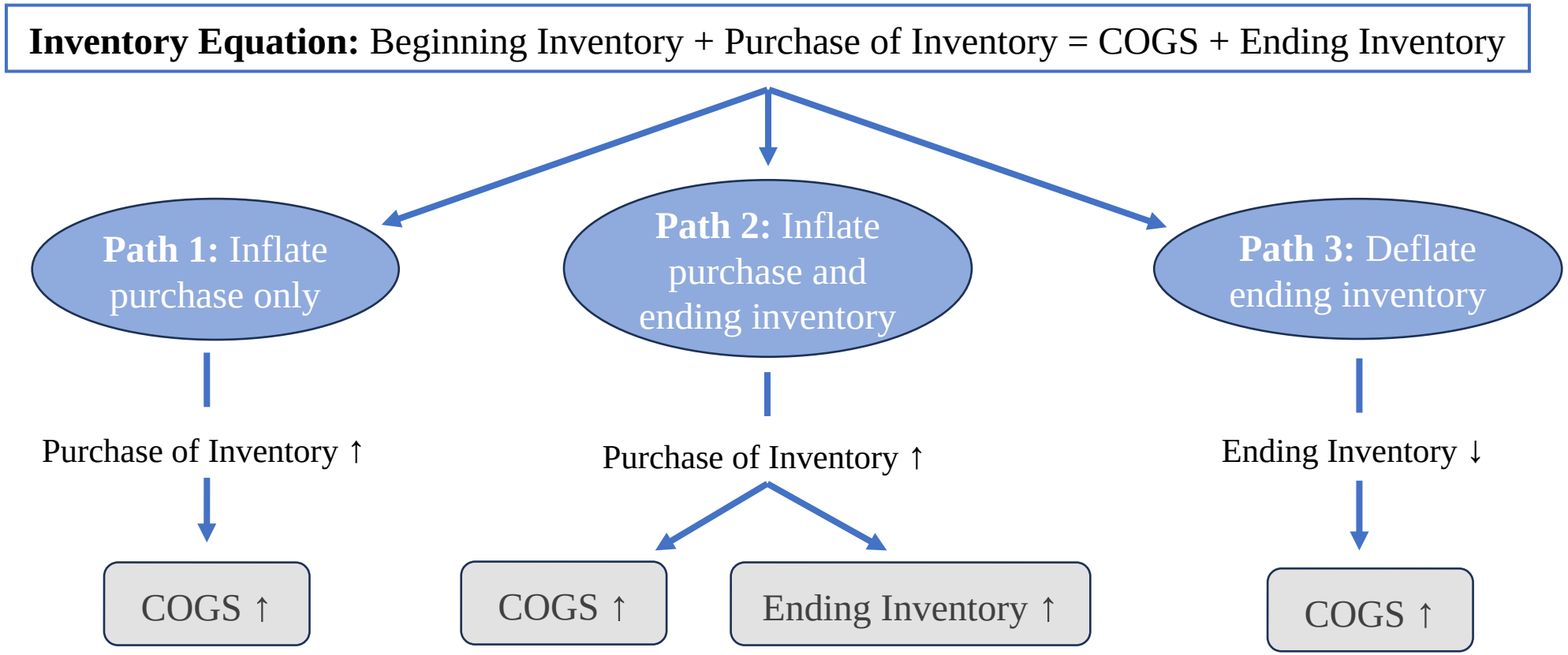
Flow Chart: Cost-Inflating Fraud Pattern

Original illustration from lecture slides:



Flow Chart: Cost-Inflating Fraud Pattern

Reconstructed mechanism:



Final Assessment of Fraud Patterns

- **Simple** fraud pattern → clearly supported
- **Complex** fraud pattern → supported, with stronger evidence from Other Receivable
- **Cost-inflating** pattern → supported, with stronger evidence from Inventory-related indicators

- Overall → Xinyan shows a **combination of multiple fraud patterns** during 2017-2019

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Thank you for listening!

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